INCH-POUND

MIL-DTL-8815/19
W/AMENDMENT 1
03 June 2025
SUPERSEDING
MIL-F-8815/19 (AS)
14 October 1976

DETAIL SPECIFICATION SHEET

FILTER ELEMENTS, NONCLEANABLE, FLUID PRESSURE, HYDRAULIC LINE, 5 MICRON ABSOLUTE, A4 AIRCRAFT, 2.0 GPM

Inactive for new design after 11 September 1996.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-8815.

REQUIREMENTS:

Envelope. See figure 1.

Pressure drop. Maximum pressure drop at rated flow across the filter element shall not exceed 10 psid.

Degree of filtration. The filter element shall remove 100 percent of all contaminants larger than 5 microns and 95 percent of all test contaminants. The filter efficiency shall be a minimum of 95 percent.

Contaminants. See table I.

Element collapse. The filter element and element end cap shall withstand a differential pressure of 750 psid.

Cold start. The element shall withstand 500 psid differential pressure across the element for the flow cycles and fluid temperature as specified in the general specification. The element pressure drop shall not exceed 10 psid following the cold start test.

Maximum particle passed. The maximum particle passed shall be not greater than 5 microns.

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Flow fatigue. The filter element shall withstand the following flow fatigue pressure cycles at a cycling rate of 100 cycles per minute (cpm) maximum.

90,000 cycles at 20 psid pressure drop
8,000 cycles at 50 psid pressure drop
2,000 cycles at 100 psid pressure drop
100 cycles at 500 psid pressure drop

Part number. The part number shall consist of the basic number of this specification sheet.

Example:

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TABLE I. Contaminants.

Test Flow Rate (GPM)	Degree of Filtration Add Rate		Maximum Particle Add Rate		Pressure Buildup		Migration Analysis Value
	F-9 Beads (grams)	AC Dust (grams)	F-9 Beads (grams)	Carbonyl Iron E (grams)	Add Rate AC Dust (grams)	Minimum AC Dust to 90 psid (grams)	Maximum Allowable (grams)
2.0	0.070	0.070	0.020	0.010	0.25	1.1	0.002

Intended use. The filter element covered by this specification sheet is intended for use in the model A-4 aircraft operating with hydraulic fluid conforming to MIL-PRF-5606 at nominal operating pressures of 3,000 psi.

Referenced documents. This document references the following:

MIL-DTL-8815
MIL-PRF-5606
SAE-AS28775

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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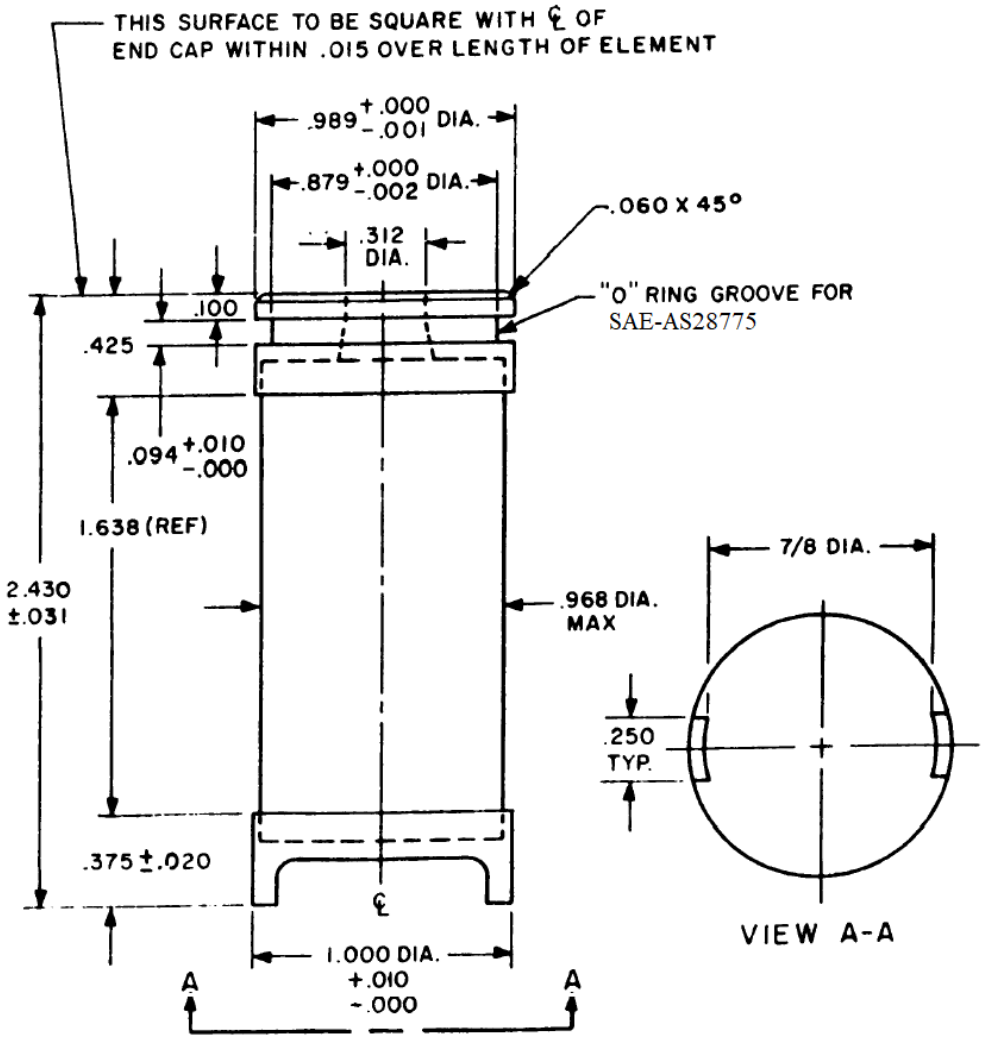


FIGURE 1. Filter dimensions.

NOTES:
1. Dimensions in inches, unless otherwise specified. See above figure 1.
2. Tolerances, decimals ±.010, angles ±l degree.
3. Unless otherwise specified, surface roughness shall not exceed 125 microinches. Remove all burrs and break sharp edges .005 minimum.
4. Material shall be in accordance with the procurement specification.

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CONCLUDING MATERIAL

Custodians:
Navy – AS
DLA – GS

Preparing activity:
Navy – AS

(Project: 1650 - 2025 - 003)

Review activities:
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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